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# Kinematic and dynamic validation of OpenCap motion capture system through Xsens IMUs

## SPORT BIOMECHANICS AND EVALUATION OF HUMAN PERFORMANCE: PROJECT

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**Abstract:** Since the human motion analysis has become more and more used in fields that are not directly related to the clinical one, it is necessary to find more intuitive and user-friendly acquisition systems as OpenCap.

The main objective of this work was to validate the OpenCap motion capture system comparing the acquisitions' results with the Xsens' ones, a leading provider of inertial sensors. The validation was conducted both from a kinematic and a dynamic point of view.

The kinematic analysis, performed through the use of MAE and ROM, led to a positive correlation between the systems' acquisitions in terms of knee and ankle flexion angles, while hip angles showed insufficient results.

From a dynamic point of view, after the computation of GRF profiles, the analysis of jump height's MAE showed comparable results between OpenCap and Xsens systems. Furthermore, an evaluation of the relationship between height of the jump and different parameters had been conducted, bringing to the conclusion that Peak Power was the most affective one.

**Key-words: OpenCap, Xsens, motion analysis, kinematic**

# 1. Introduction

Nowadays the field of human body motion tracking is becoming more and more relevant with the addition of new research activities. The studying of dynamics and kinematics of human body is fundamental to understand the behavior of musculoskeletal and neuromuscular systems in medical, research and high-performance applications [1]. The term motion capture (MoCap) can have different meanings, based on the several application areas. MoCap can relate for example to the recording of the movement of objects or people. Particularly, it is possible to classify motion capture systems in two classes: marker-based and marker-less systems [2].

Literature highlights marker-based systems as the best in both accuracy and efficiency for movement analysis [3] - [6]. Indeed, in the last 20 years, motion capture technology has significantly improved: cameras' resolution is continuously evolving, reaching over one megapixel (MP), and tracking software is getting improved day-by-day, allowing marker replacement/switching, gap filling, and signal filtering [7].

However, marker-based systems could be very expensive, even considering different constraints related to the environment and the subject's preparation [1]. A fully equipped lab could reach investments of thousands of dollars, with several days of work to collect and analyze data (Figure 1). Moreover, using marker-based systems during high performance competitions and sport events is difficult and uncomfortable for the players due to the presence of trackers.

Thus, a relevant compromise has been found in Inertial Measurement Units (IMUs), which allow measurements of kinematics outside the lab, preserving the relationship athlete-environment [8]. IMUs are placed on body segments, and with them it is possible to determine the global position, angular velocity, acceleration, and other parameters for each of the segments [9].

However, in recent years efforts have been concentrated to develop suitable marker-less systems for motion capture, easier for users to understand and to exploit all measurements [5]. Indeed, this type of MoCap systems are more *consumer friendly* because they do not require time and labor to process data (Figure 1). Nonetheless, these advantages in accessibility and application environments could reduce measurement accuracy, leading to uncertainties and errors in the reconstruction of the movement.

For the reasons presented, the aim of this study is to compare data acquired with OpenCap system, based on two smartphone cameras, with a gold-standard system of motion analysis, based on X-Sens IMUs, through the evaluation of several simple exercises.

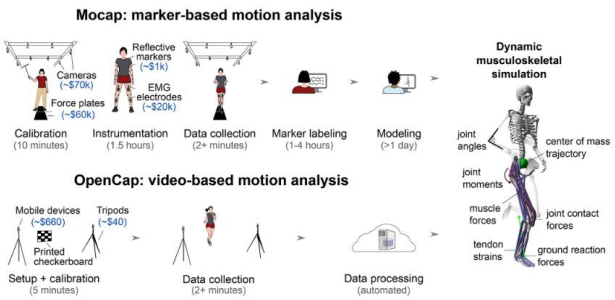


*Figure 1. Marker-based motion capture (Mocap) versus video-based (OpenCap) analysis of human movement dynamics: fundamental steps, related costs and labour-times* [1].

# 2. State of Arts

## 2.1 Optoelectronic systems

Optoelectronic systems are the gold standard for motion capture and analysis in various fields, particularly in biomechanics, sports science, and rehabilitation. This system offers high precision and non-intrusive measurements, for a wide range of applications.

Optoelectronic system is composed of a multiple set of cameras that emit infrared (IR) light, invisible to the naked eye. Reflective markers are attached to specific points of interest on the subject's body (e.g., joints, limbs); they reflect the IR light emitted by the cameras, allowing the capture of the position of the markers within its field range. Finally, a specialized software is required to track the positions of the markers in each camera frame, reconstructing the 3D movement of the subject and generating its digital representation. The advantages of Optoelectronic Systems are:

- High precision: high level of accuracy for complete analysis of movements.
- Non-intrusive: IR light doesn't affect subject's motion, unlike wired sensors.
- Large capture volume: cameras can capture movement across a wide area.
- Data adaptability: captured data are useful for various analyses, including kinematics (focusing on joint angles), kinetics (forces and moments evaluation), and gait analysis.

On the other side, there are some disadvantages for Optoelectronics Systems, like:

- Cost: initial investment for camera systems and software is very expensive.
- Setup Time: calibration and placement of markers on the subject is time-consuming.
- Limited movements: markers must be detectible by cameras for accurate tracking; this could be limiting, for example for complex movements or different environments analyses.
- Limited environment: optoelectronic system, with the cameras set-up, is difficult to carry out from laboratory in different locations, limiting the applications.

Optoelectronics system is fundamental in biomechanics application, both for studying human movement patterns and mechanics, efficiency, and preventing injuries in sport science application, for analyzing high level performances and optimizing sport-specific techniques, and in rehabilitation, for monitoring patient's progress in physical therapy and rehabilitation programs (and in other several applications).

Optoelectronic systems remain probably the best between acquisition systems for MoCap due to their high precision and versatility. Otherwise, it is required a larger investment and setup time compared to some alternatives; they are still considered the gold standard for detailed motion analysis in various fields.

## 2.2 Inertial sensor measurement systems (IMU)

A valid alternative to Optoelectronic Systems is the *Inertial Measurement Units* (IMUs).

IMUs are wireless and wearable sensors that track movement. They offer several advantages, particularly in terms of portability and ease of use. IMUs are compact devices that measure and report an object's motion and orientation. Typically, they are composed by an accelerometer for measuring linear acceleration along multiple axes, gyroscope for the evaluation of angular velocity around multiple axes and often, a magnetometer, for the detection of the ambient magnetic field, helping in orientation tracking. In addition, it's necessary to integrate internal algorithms, called *sensor fusion*, to combine and process the raw data measured by those devices. After an initial

calibration, the IMU estimates its position and orientation over time based on the measured accelerations and rotations. The advantages of IMUs for MoCap are the portability and the light weight, since the devices are small and wireless, and they can be easily attached to the body without hindering movements. Furthermore, IMUs are easy to set up, because they do not require a complex camera system nor a controlled environment. In the end, their cost is lower compared to optoelectronic systems and their applicability is wider; the system is suitable for capturing movement outdoors or in open environments.

Otherwise, IMUs' data tends to drift over time due to sensor errors and integration inaccuracies. This can lead to the addition of errors in the reconstructed movement, especially during long recordings. Generally, IMUs are considered less precise than optoelectronic systems, especially for complex movements or fast rotations analyses, and a proper calibration is crucial to reach precise results, requiring specific trials. Eventually, metal objects or magnetic fields could interfere with the magnetometer, altering orientation tracking.

While IMUs offer significant advantages in terms of portability and ease of use, their accuracy limitations need to be considered. As sensor technology improves and data processing techniques advance, IMUs are poised to play an increasingly important role in MoCap, particularly in applications where portability and affordability are crucial factors.

# 3. Materials and Methods

## 3.1 Experimental sample

The experimental sample was composed by 15 subjects, different in gender, height, weight, level of wellness and strength. The height range is from 164 cm (5 ft 4 in) to 185 cm (6 ft 1 in), while weight range goes from 55 kg (121 lbs) to 96 kg (212 lbs). All the individuals are young adults, between 22 and 24 years old. Furthermore, 4 out of 15 individuals have a history of injuries or surgeries, including knee and Achille's ligament injuries.

## 3.2 Inertial measurement units (IMUs) - Xsens

Xsens' IMUs were used as gold standard to compare and validate the marker-less motion capture system OpenCap. This type of wireless motion tracker (Xsens MTw Awinda-16g) is provided with 3D linear accelerometers, 3D rates gyroscopes, 3D magnetometers and a barometer, that makes it suitable for a fast and simple recording of the several body segments' movements. The sensors are also provided with different stations to charge and connect them to the software (Xsens Analyze) used to build the kinematic model of the subject under observation. Moreover, strip bands and special t-shirts are included with the trackers to facilitate the wearability.

For the case in study, a total of fifteen IMUs were used on the subjects, as suggested by one of the models implemented in the Xsens software (full body-no hands). Particularly, the trackers were placed on the forehead, stern, right/left shoulder, right/left forearm and upper arm, pelvis, right/left upper leg and lower leg and on the two feet. The most relevant sensors were the ones applied on the lower limbs to evaluate the kinematic of hip, knee, and ankle joints. The sampling frequency used for each tracker was set to 60Hz due to limitations of the wireless transmission of data. A scheme of the model used for the Xsens system is shown in the Figure 2.

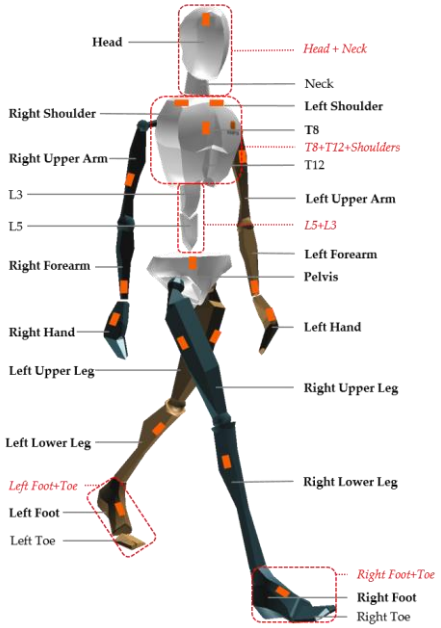


*Figure 2. Xsens IMUs' model used for experimental acquisitions. It is a full body scheme, and in the performed tests the hands' IMUs weren't used.*

## 3.3 OpenCap

OpenCap, is a web-based software that is freely available to the research community for estimating the 3D kinematics and dynamics of human movement from videos captured with two or more smartphones. OpenCap makes the analysis of movement dynamics available without specialized hardware, software, or expertise. OpenCap set-up took about five minutes and required two (or more) iOS devices (in this work two iPhones have been used), two tripods (placed at less than 5 meters from the recorded volume and with an orientation angle between 30° and 45° with respect to the frontal orientation of the recorded volume), a printed calibration checkerboard and a further device to run OpenCap's web application (a laptop). After pairing the iOS devices, it was necessary a camera calibration (performed with the previous cited checkerboard). Then it was possible to acquire data at a sampling frequency of 60 Hz with OpenPose and HRNet modalities (for the case in study OpenPose modality was used), and to visualize the 3D kinematics. Kinematics were estimated from video using deep learning models and inverse kinematics in OpenSim, while dynamics were estimated using a physics-based musculoskeletal simulation approach [10]. OpenCap leverages cloud computing for data processing using a scalable server architecture.

## 3.4 Protocol

The acquisitions in the laboratory started with the setting of the two systems used for the movement's analysis. Firstly, all the inertial sensors were placed on specific segments of the subject's body (as shown in the previous section). Then, the subject was asked to keep a standing position in front of a given paper grid for the calibration of the cameras used for the OpenCap and immediately after to pass to the calibration of the Xsens system (3s standing position, 10s walking and stance for further 10s). Once this step returned a positive result, the subject had to perform three consecutive repetitions of a squat (SQ), that were recorded in a single acquisition. Following, three more trials of a squat jump (SJ) were acquired separately one from the others, and the same protocol was taken for a drop jump (DJ) from a wood box of 30 cm of height. During all the three types of exercise the

subject was asked to keep the hands in contact with the hip to have standardized movement among the fifteen participants (Figure 3). At the end of the experimental laboratory, seven acquisitions per subject were recorded (one for SQ, three for SJ and three for DJ), leading to a total of 105 data history both for the OpenCap and Xsens systems.

*Figure 3. Setup of the experimental laboratory, composed by Xsens IMUs, two cameras and paper grid for the OpenCap.*

# 4. Results

## 4.1 Analysis of kinematics

The kinematic analysis of the OpenCap system respect to the Xsens, assumed as gold standard, was performed by comparing three main angles: hip, knee and ankle flexion/extension angles both for right and left limb in the sagittal plane (Figure 4). The choice was based on the greater mobility of these joints and on the certainty that the angles acquired by the different systems were comparable. Several methods to compare the two systems were chosen, but firstly the *mean absolute error* (MAE) and the *range of motion* (ROM) approaches.

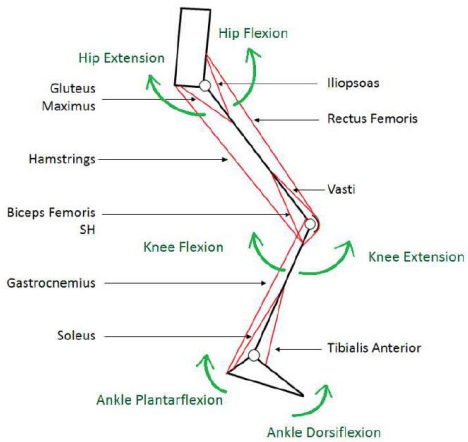


*Figure 4. Angles of interest in the sagittal plane (green ones)*

The comparison was evaluated on the three types of exercises proposed (squat, squat jump, drop jump) and the acquisitions were performed as explained in the protocol of the experimental laboratory. It must be highlighted that between the OpenCap and Xsens acquisitions a general delay was always occurring because of the uncertainty related to the different operator, and the one of the cameras and IMUs promptness.

After all the recordings were completed, the data obtained with the two systems were analyzed and compared with the use of MATLAB®. The first step consisted in the evaluation of how the several exercises were performed by the different subjects: for each protocol the 45 trials acquired with Xsens (gold standard) were overlapped and the mean value, standard deviation and the *root mean square* (RMS) were calculated (Figure 6). As it's possible to see in the example of Figure 5 the values over time had a wide deviation due to the different speeds of the subjects to perform the exercise, which were not imposed as fixed during the protocols. However, this consideration did not affect the results for the validation of the OpenCap because the comparison was done within the same subject and between the two different systems. A table that resumes the average maximum and minimum values of the range of the different angles is also shown below (Table 1).

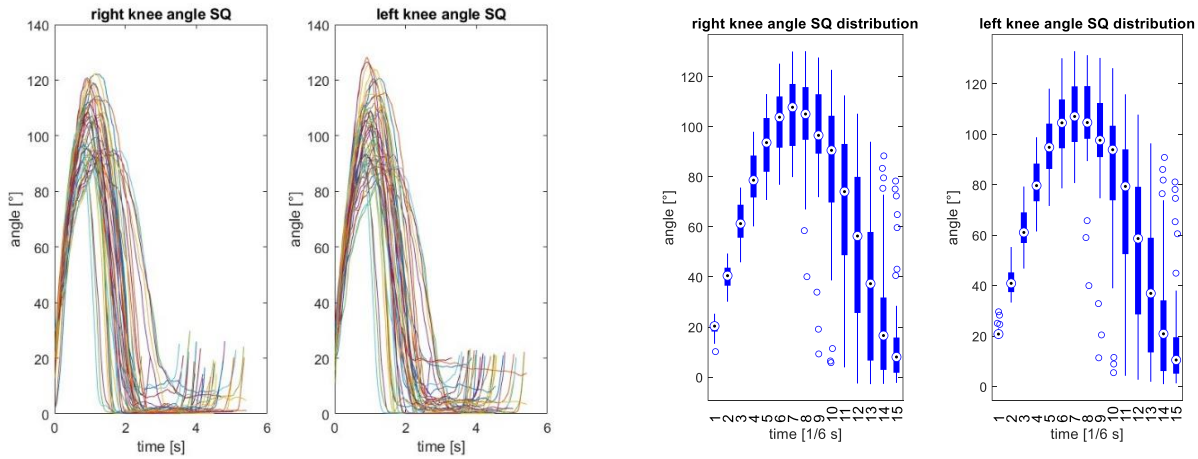


*Figure 5. Right and left knee angles during different SQ trials (left) and their distributions around the mean value (right)*

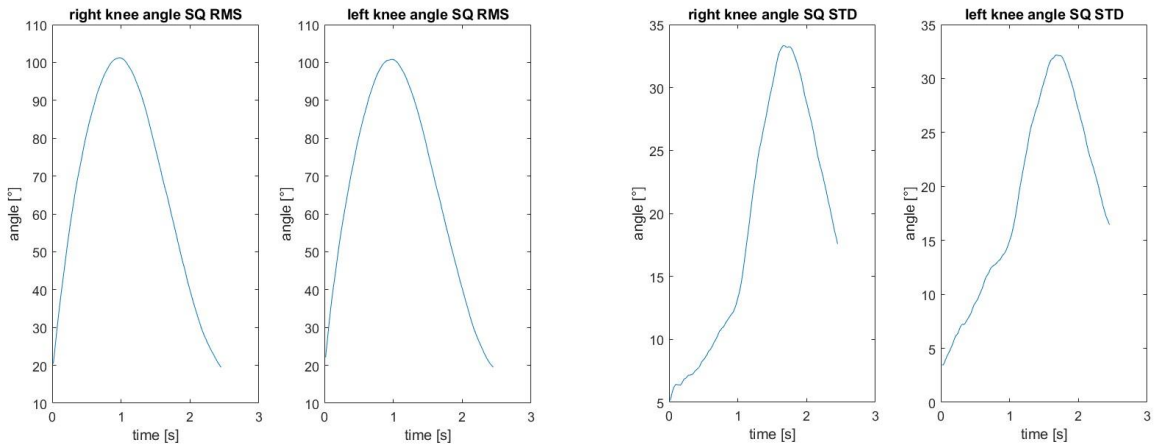


*Figure 6. RMS and STD about the SQ exercise*

*Table 1: Angles' range*

| Angle range | SQ [°] | SJ [°] | DJ [°] |
|---|---|---|---|
| **Knee R** | 106.5-22 | 97.7-21 | 75.3-20.6 |
| **Knee L** | 108.6-21.6 | 99.6-23.0 | 75.5-22.2 |
| **Ankle R** | 28.5-11.6 | 31.6-11.1 | 29.7-11.6 |
| **Ankle L** | 31.1-13.1 | 31.9-12.5 | 28.9-9.9 |
| **Hip R** | 125.8-20.4 | 115.6-20.1 | 90.5-37.7 |
| **Hip L** | 127.7-22.8 | 117.8-20.8 | 92.0-37.5 |

Moving to the kinematic comparison of the two systems based on the MAE approach, the first problem encountered was the shift of angle's value between the OpenCap and the Xsens, probably related to the different way of calibration and the wrong positioning of the trackers detected by the software. In some cases, mainly with the OpenCap system, the starting angle was not realistic, with a translation of the curve upward or downward for all the angle values, leading obviously to an impact on the computation of the errors. Examples related to the drop jump trial of one subject are shown in Figure 7,where it is possible to see that even if the movement was the same the initial values on the vertical axis were different. Two methodologies were adopted to validate the marker-less system, varying in the way to align the signals in time and in the consideration or not of the vertical shift.

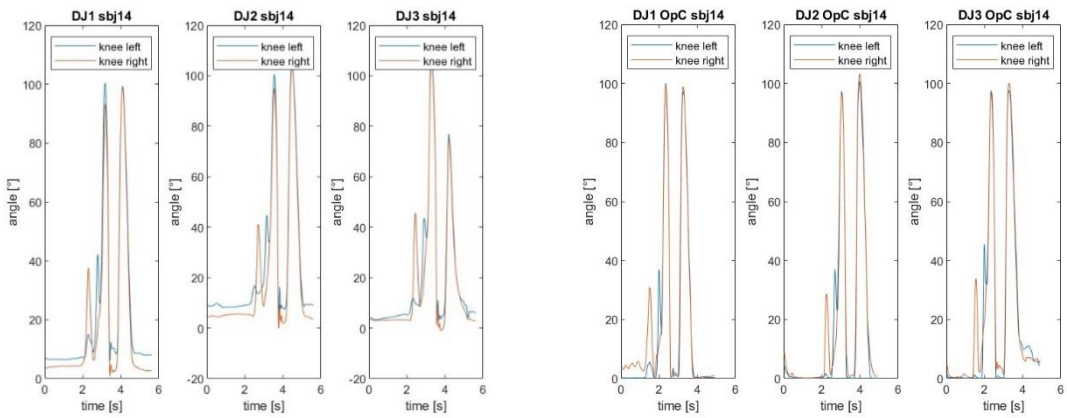


*Figure 7. Drop jump trial for subject 14 recorded by Xsens (left) and OpenCap (right)*

### 4.1.1 Alignment by starting angle value and MAE computation

The first method relied on the alignment of the same OpenCap and Xsens' signals through the manual definition of a specific threshold, used as starting point in order to have the same value of reference for the two curves and possibly the alignment. The vertical shift in this case was not

corrected. The next step was the cut of the signals tails to have congruent dimensions between them in order to compute instant per instant the Mean Absolute Error ($MAE = \frac{\sum error\ (t_i)}{n}$), used based on what found in the literature [1]. This parameter is a measure of the average size of the mistakes in a collection of two data series. It is measured as the average absolute difference between the first value and the second one and it is used to assess the effectiveness of a model or in this case the reliability of a system of acquisition.

The MAE was computed for each trial of all the exercise and repeated for all the subjects, in order to obtain at the end a mean value of the parameter for every angle associated to a specific protocol. In Figure 9 an example of the MAE and the relative percentage error (Figure 8) calculated for every time instant are reported.

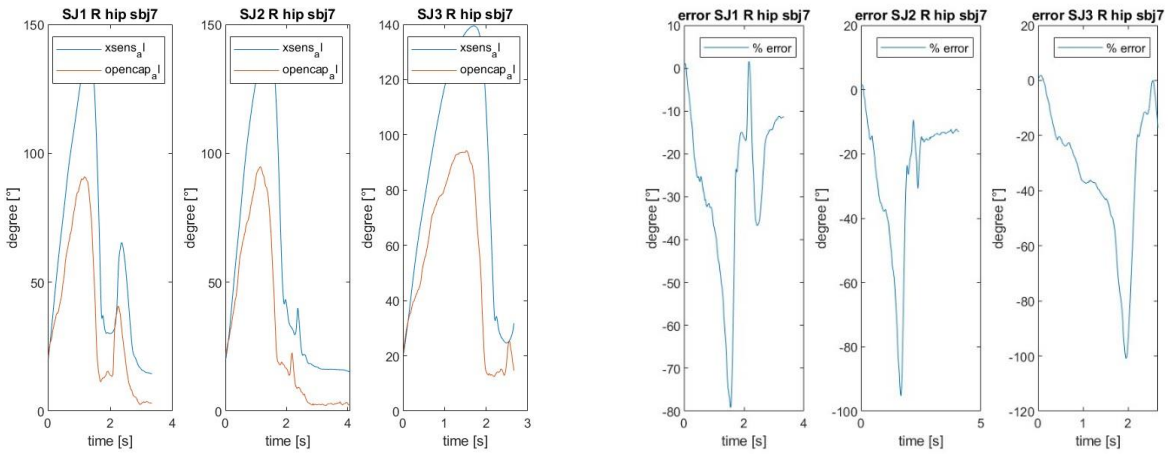


*Figure 8. Aligned signals (left) and errors (right) in the case of SJ's trials for Subject 7*

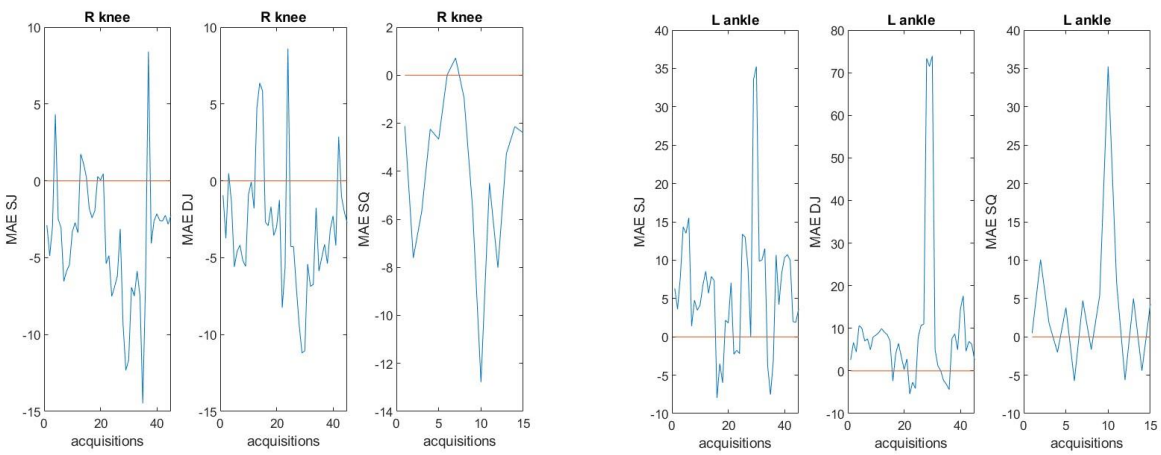


*Figure 9. MAE values (on all the acquisitions and for the 3 exercises) for the knee angle (left) and ankle one (right)*

Eventually, the mean and the standard deviation of the MAE were computed by considering the 45 available values of it, gotten by the three trials per subject of SQ, SJ and DJ. The parameters were evaluated for all the angles analyzed (right and left hip, knee, ankle flexion/extension), and they are

represented in the following Figure 10, as their exact values are reported in Table 2 and Table 3.

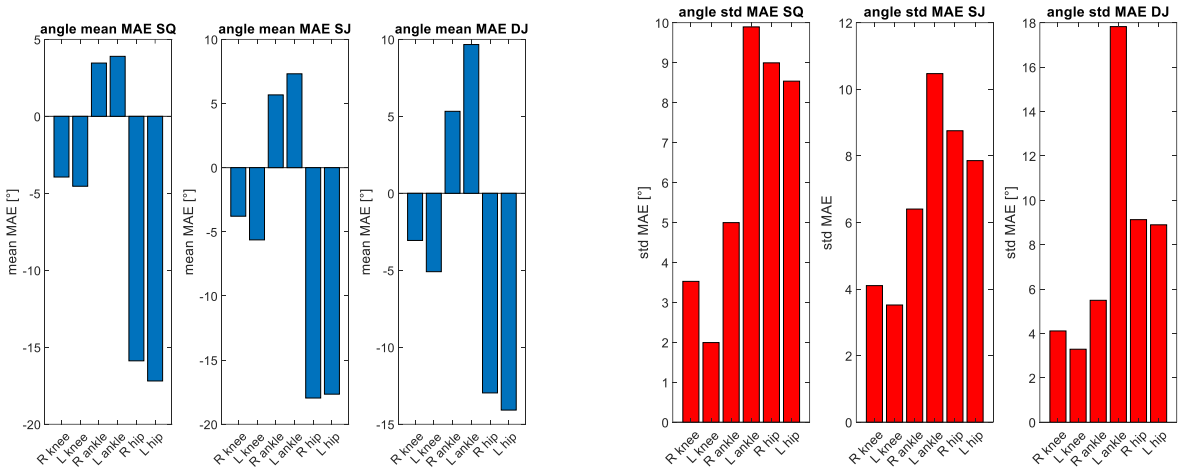


*Figure 10. Mean values (right) and STD (left) of the MAE for all the investigated angles*

*Table 2: Mean MAEs*

| Mean MAE | SQ [ ° ] | SJ [ ° ] | DJ [ ° ] |
|---|---|---|---|
| **Knee right** | -3.9427 | -3.7857 | -3.0625 |
| **Knee left** | -4.5417 | -5.6310 | -5.0914 |
| **Ankle right** | 3.4609 | 5.6714 | 5.3258 |
| **Ankle left** | 3.8968 | 6.2539 | 9.6624 |
| **Hip right** | -15.8818 | -17.9568 | -12.9656 |
| **Hip left** | -17.1898 | -17.6562 | -14.0803 |

*Table 3: MAEs' STDs*

| Std MAE | SQ [ ° ] | SJ [ ° ] | DJ [ ° ] |
|---|---|---|---|
| **Knee right** | 3.5258 | 4.1032 | 4.1131 |
| **Knee left** | 1.9939 | 3.5216 | 3.2897 |
| **Ankle right** | 4.9960 | 6.4030 | 5.4934 |
| **Ankle left** | 9.8948 | 8.5824 | 17.8238 |
| **Hip right** | 8.9949 | 8.7547 | 9.1267 |
| **Hip left** | 8.5370 | 7.8576 | 8.8942 |

The results showed errors that in many cases cannot be completely neglected, particularly in the detection of the hip joint, where a critical situation of the average error that exceeded 10° was not negligible. The other values of the MAE were quite stable, with values around 5° of error. Nonetheless, it could be observed that the knee was the angle where the OpenCap system works as best in accordance to the Xsens, because the MAE values showed a limited error with a standard deviation of only few degrees. On the other hand, for the ankle and firstly for the hip the results were worse, considering also the standard deviation.

### 4.1.2 Alignment by peaks and MAE computation

The second method that was adopted to compare the angles' acquisition of OpenCap with the one of Xsens relied on the synchronization of the two signals by means of MATLAB® implemented functions. The following procedures were applied to the most relevant angles (right and left hip, knee and ankle flexion/extension) for each of the fifteen subjects. As first step of the analysis, considering subject by subject, the curves of each exercise (SQ, SJ and DJ) were manually aligned in terms of starting angle (stance phase), due to a different calibration of the two systems, in order to have a correct evaluation of the error to validate the marker-less instrumentation. Consequently, a different approach was used for the squat with respect to the SJ and DJ to align the curves acquired with the two systems. This differentiation was related to the fact that all the three repetitions were registered with one single acquisition for the squat, while three separated trials were recorded for the SJ and DJ. Concerning the squat, the single time history of the angles was separated squat by squat identifying the three peaks of flexion with the function *findpeaks()* both for OpenCap and Xsens; then, a sufficient and constant number of points was considered to cut the actual time of exercise starting from the detected peaks of each trial, and the alignment in time of all the curves was performed by means of the same peaks. On the other hand, the squat jump and drop jump angles' acquisitions were aligned for each trial using the MATLAB® function *alignsignals(),* which is based on finding the best cross-correlation between two curves; a second alignment was performed with the same function between the three couples (one for each trial) to be able to cut all the curves in a common range of constant length. After a visual check of the positive result, the curves were cut as in the squat case by taking the correct number of points to define the exercise phase. As a final result, the signals of right and left hip, knee and ankle flexion coming from OpenCap and Xsens were obtained as correctly aligned for all the subjects, as shown in Figure 11, Figure 12 and Figure 13.

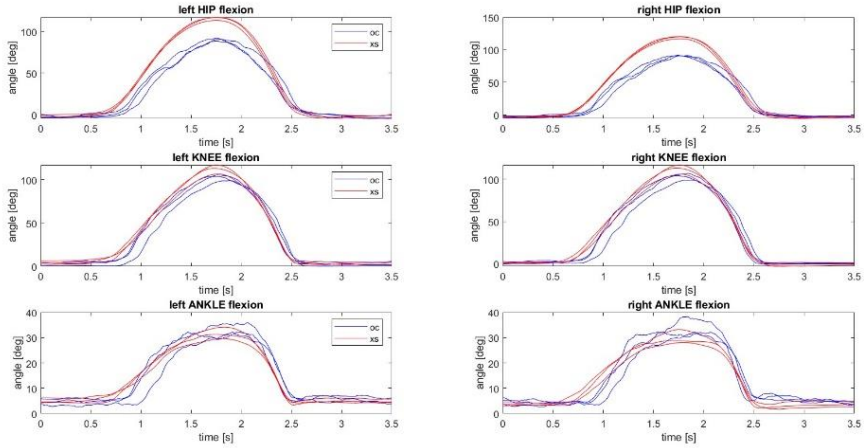


*Figure 11. Alignment of all the acquisitions (for one subject) for the squat, with OpenCap (blue) and Xsens (red).*

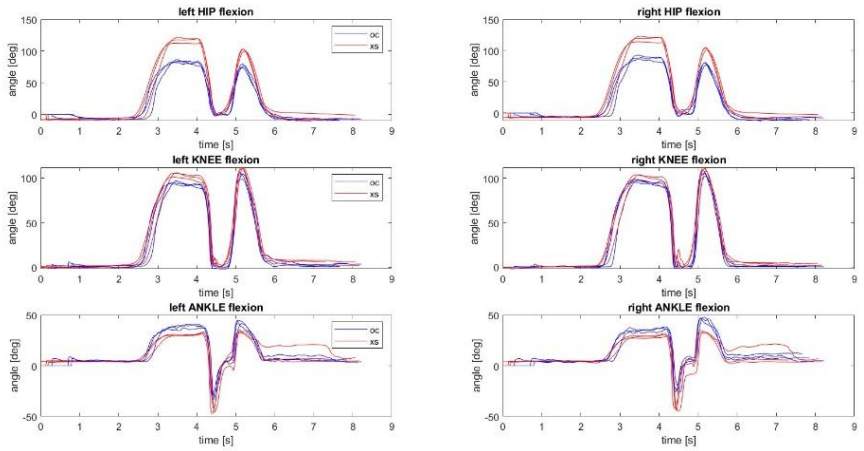


*Figure 12. Alignment of all the acquisitions (for one subject) for the squat jump, with OpenCap (blue) and Xsens (red).*

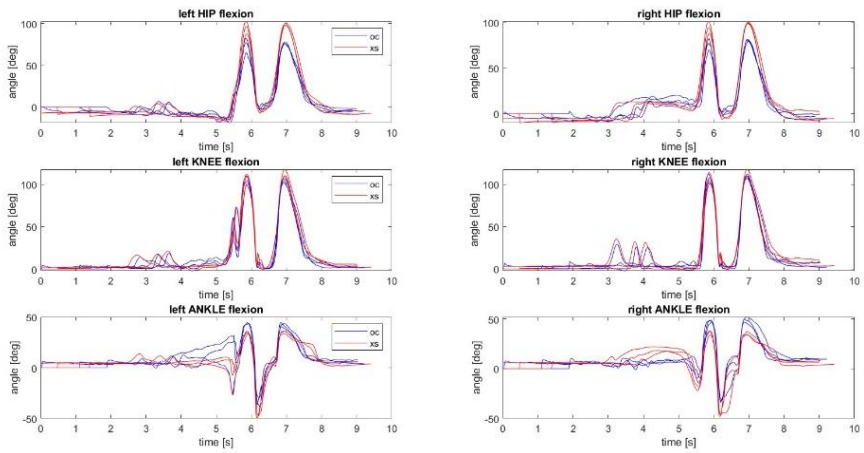


*Figure 13. Alignment of all the acquisitions (for one subject) for the drop jump, with OpenCap (blue) and Xsens (red)*

After the correct alignment and cut of the signals, the mean absolute error (MAE) between the two systems was calculated for all the angles to evaluate the reliability of OpenCap in acquiring the kinematic. The parameter was obtained by taking the absolute error ($|angle_{opencap} - angle_{xsens}|$) in each time instant and by performing a mean of it. This process was done for all the three trials of the exercises, considering the fifteen subjects. The results for SQ, SJ and DJ are shown respectively in Figure 14, Figure 15 and Figure 16.

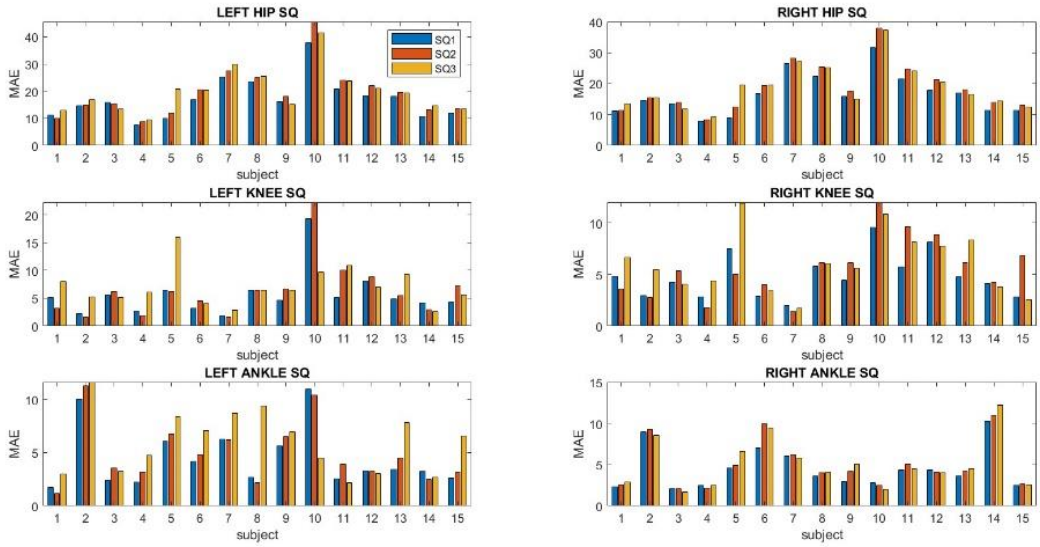


*Figure 14. Mean absolute error (MAE) calculated for each trial of the squat, for all the subjects.*

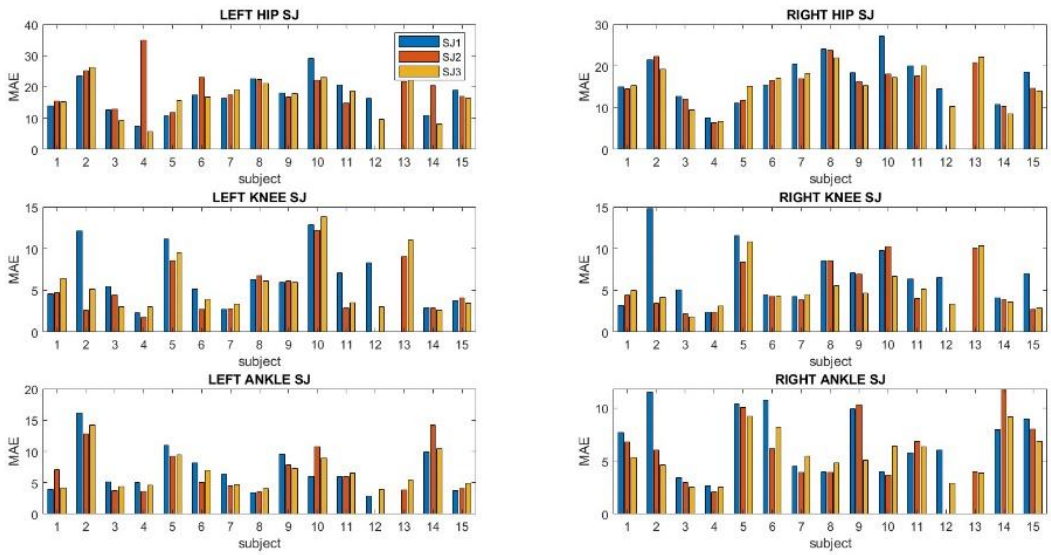


*Figure 15. Mean absolute error (MAE) calculated for each trial of the squat jump, for all the subjects.*

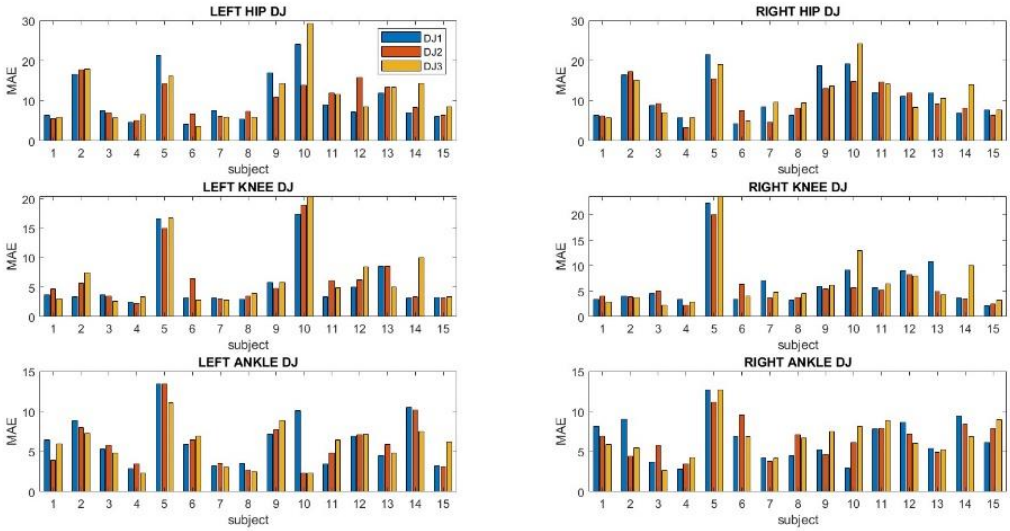


*Figure 16. Mean absolute error (MAE) calculated for each trial of the drop jump, for all the subjects.*

As last step of the analysis, the mean and standard deviation of the MAE were calculated for each type of protocol, based on a set of 45 values (three trials for fifteen subjects). As in the first method, only the squat jump relied on a lower number of data (43), due to missed acquisitions of the OpenCap system for subject n°12 (second trial) and n°13 (first trial). The results are reported in Figure 17.

It was possible to see that a mean absolute error of 5° was present in the different acquisitions of the knee and ankle angles, which could be considered as a reasonable result for a marker-less system compared to a gold standard. The range of variation of this error was common between the three types of squats leading to affirm that its value was not dependent on the category of exercise or movement performed. However, it must be highlighted that a bigger error of around 20° has been found for the hip flexion/extension comparison. This increase in the difference could be caused by the more complicated process of recognition of the hip joint by OpenCap, that had to face the disturbance of clothes (t-shirt and shorts), differently from the knee and ankle joints that were more visible. In conclusion, the results gained with the method just shown were correctly in line with the ones previously described with the first approach.

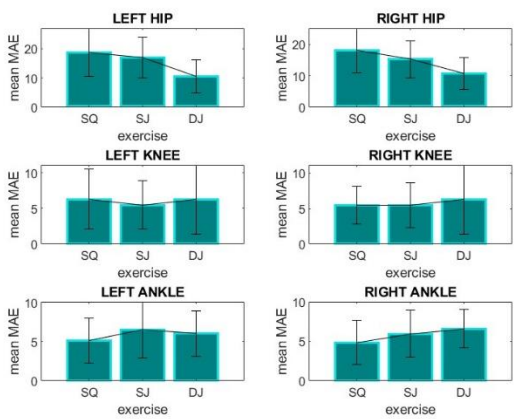


*Figure 17. Mean and std of MAE calculated for each type of exercise (squat, squat jump, drop jump).*

### 4.1.3 Range of motion

As last point, the validity of the OpenCap was verified in terms of the range of motion (ROM) recorded for the different joints. This approach was adopted to find results independent from the different starting angles of the two systems, because based on the consideration of the distance between two peaks' values (maximum and minimum). However, it must be appointed that this methodology was more susceptible to errors related to abnormal peaks for the movement detection in particular time instants. The ROM, calculated as the difference between the maximum and minimum value of the angle of each specific joint, was extracted from all the trials of the fifteen subjects for each of the three exercises. Then, mean, standard deviation, kurtosis, and skewness of the range were computed for the SQ, SJ and DJ, and reported in Table 4, Table 5 and Table 6. In Figure 18 are shown the different values of ROM for the SJ coming from Xsens and OpenCap.

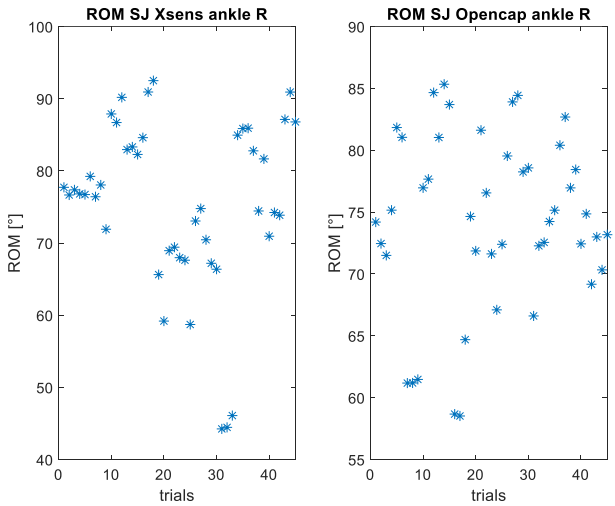


*Figure 18. Ankle's ROM data of SJ of all the trials for Xsens (left) and OpenCap (right)*

*Table 4: Statistic coefficients for SQ (Xsens – OpenCap)*

| Statistic coefficients SQ | MEAN [°] | STD [°] | KURTOSIS [-] | SKEWNESS [-] |
|---|---|---|---|---|
| **Knee right** | 109.1 – 103.5 | 13.1 – 10.5 | 2.3 – 2.0 | 0.3 – 0.01 |
| **Knee left** | 108.2 – 101.9 | 13.3 – 11.4 | 2.3 – 2.3 | 0.4 – 0.6 |
| **Ankle right** | 27.3 – 29.1 | 4.4 – 8.1 | 2.6 – 2.5 | (-0.2) – (-0.6) |
| **Ankle left** | 28.6 – 30.3 | 8.0 – 7.6 | 2.9 – 2.9 | 0.2 – (-0.4) |
| **Hip right** | 130.0 – 105.4 | 13.3 – 3.8 | 3.1 – 2.1 | 0.9 – 0.7 |
| **Hip left** | 131.2 – 106.1 | 13.3 – 5.8 | 2.2 – 2.2 | 0.8 – 0.9 |

*Table 5: Statistic coefficients for SJ (Xsens – OpenCap)*

| Statistic coefficients SJ | MEAN [°] | STD [°] | KURTOSIS [-] | SKEWNESS [-] |
|---|---|---|---|---|
| **Knee right** | 111.0 – 106.1 | 12.9 – 10.7 | 2.3 – 2.1 | 0.1 – 0.1 |
| **Knee left** | 110.9 – 104.8 | 14.1 – 12.9 | 2.4 – 2.4 | 0.4 – 0.3 |
| **Ankle right** | 75.4 – 74.3 | 11.7 – 7.1 | 3.9 – 2.7 | (-1.1) – (-0.5) |
| **Ankle left** | 69.2 – 71.0 | 14.3 – 12.5 | 2.2 – 3.7 | (-0.7) – (-1.0) |
| **Hip right** | 131.1 – 97.7 | 13.0 – 9.1 | 2.8 – 2.1 | 0.7 - (-0.03) |
| **Hip left** | 131.4 – 97.5 | 13.3 – 10.5 | 3.1 – 2.2 | 0.9 – 0.2 |

*Table 6: Statistic coefficients for DJ (Xsens – OpenCap)*

| Statistic coefficients DJ | MEAN [°] | STD [°] | KURTOSIS [- | SKEWNESS [-] |
|---|---|---|---|---|
| **Knee right** | 105.6 – 101.0 | 16.6 – 15.0 | 2.0 – 2.0 | (-0.03) – (-0.01) |
| **Knee left** | 104.6 – 99.3 | 15.9 – 15.1 | 2.1 – 2.2 | (-0.08) – 0.04 |
| **Ankle right** | 73.3 – 75.1 | 15.1 – 6.8 | 5.7 – 3.1 | (-1.5) – (-0.8) |
| **Ankle left** | 70.0 – 69.3 | 14.8 – 12.1 | 2.2 – 4.0 | (-0.7) – (-1.1) |
| **Hip right** | 104.9 – 83.1 | 24.9 – 19.7 | 2.1 – 1.9 | (-0.3) – (-0.6) |
| **Hip left** | 104.6 – 82.9 | 24.1 – 20.6 | 2.0 – 1.8 | (-0.5) – (-0.5) |

A first consideration could be done looking at the results of the standard deviation. It was possible to observe that the STD of the OpenCap was generally lower than the one of the Xsens, showing that the first acquisition system seemed less susceptible to the change of the subject during the same exercise, even though the difference of the mean values of the two systems was not so relevant. In addition, it could be seen a similarity in the results with the studies already present in literature (Uhlrich et al. 2023 [10]), in which the OpenCap seemed to record angular values lower than the motion capture system's ones. The same has been found in terms of mean value, comparing the OpenCap with the Xsens system.

The last step in the analysis of the ROM concerned the evaluation of a similar behavior of the mean value calculated by the two systems, by using statistical tests. To prove the similarity, the first passage was the definition of the distribution of the pool of data: a non-normal distribution of the data was found through the Kolmogorov-Smirnov test. The test assumes as null hypothesis that the data comes from a standard normal distribution; it returns h=1 if the hypothesis is discarded. Having considered the previous result, the two-sided Wilcoxon signed-rank test (that returns the *p*-

*value* and h) was used to study the correlation between the two paired series of data. This test assumes the null hypothesis that the data in the general vector (x-y) comes from a distribution whose mean value is zero at 1% significance level (*p-value*) respect to standard 5%: so, for each exercise, the two different series of data for each angle were taken by the test, in which h=1 indicates a rejection of the null hypothesis, while h=0 the opposite. The results of this last test and the respective *p-value* are reported in Table 7 and Table 8 for the validation by statistical analysis.

*Table 7: Statistical Validation*

| VALIDATION | SQ | SJ | DJ |
|---|---|---|---|
| **Knee right** | OK | X | OK |
| **Knee left** | OK | X | X |
| **Ankle right** | OK | OK | OK |
| **Ankle left** | OK | OK | OK |
| **Hip right** | X | X | X |
| **Hip left** | X | X | X |

*Table 8: Statistical validation (p-values)*

| VALIDATION (*p-value*) | SQ | SJ | DJ |
|---|---|---|---|
| **Knee right** | 0.2318 | 1.958e-06 | 0.0117 |
| **Knee left** | 0.3126 | 6.219e-08 | 9.923e-07 |
| **Ankle right** | 0.3369 | 0.5997 | 0.7974 |
| **Ankle left** | 0.1792 | 0.4328 | 0.4262 |
| **Hip right** | 0.0020 | 3.569e-08 | 5.234e-05 |
| **Hip left** | 0.0020 | 3.569e-08 | 6.948e-07 |

Hence, from the results it was shown that the number of situations in which the OpenCap and Xsens were correlated was half of the cases. In particular, the hip joint presented the worst case with no correlation detected between the two systems. This problem could be linked to the difficulty of the OpenCap to find the right joint position due to the clothing and the position of the cameras, as previously supposed in the MAE analysis. Concerning the other outcomes, it was possible to see that the ankle joint was the best correlated with the Xsens and the squat exercise presented a better overall result than the other two protocols, probably because based on a slower and more controlled movement. In conclusion, it must be appointed that a higher correlation of the OpenCap with the gold standard was expected, but probably the use of inertial sensors rather than a motion capture

system introduced wider ranges of error due to the positioning of the IMUs and their displacement caused by soft tissues.

## 4.2 Analysis of Dynamics and Performances

Since the analysis of dynamics and of the performances is based on forces estimations, the evaluation of *ground reaction forces* (GRF) is fundamental: by definition they are the forces exerted by the ground on an object in contact with it, and in biomechanics this object is the human body. For the estimation of the GRF profiles and for the extraction of the most important parameters several passages had to be completed, starting from the acquisitions performed both with OpenCap and Xsens. To better describe all the procedure followed to arrive to the results this second part of the report is divided in different sections, which are presented in the successive paragraphs and where each passage (for each single acquisition) is described.

Since the two acquisition systems used in the protocol were not equipped with force plates, the unique method to evaluate the GRF generated by the athletes in the time instants related to the jump were the ones based on the tracking of the positions of the CoM of the body segments. Indeed, by extracting these markers positions, by computing their velocities and their accelerations, and by taking into consideration their proper segment's masses, the estimation of the profile of the GRF for each recorded time instant was possible.

### *4.2.1* Extraction of markers' positions, adoption of the same reference system and synchronization

The first fundamental task to be performed was the extraction of the anatomical body markers' positions. This had to be done both for the signals acquired with OpenCap and with Xsens. From each subject's downloaded files, the positions of each single marker for each single time instant of the acquisitions were extracted and saved in MATLAB® for further analyses. OpenCap acquisition system referred on and managed to extract the positions of 63 anatomical body markers for each single time instant of acquisition, instead Xsens gave as output as *Segment Positions* the placement in the space of 23 points that were the schematic representation of the CoMs of body's segments. From their positions, both absolutes (with respect to an absolute reference system) and relatives, proper and dedicated software managed to extract an anatomical model. In Figure 19 the models recreated for OpenSim and Xsens from the acquired body segment positions and with the proper software are presented.

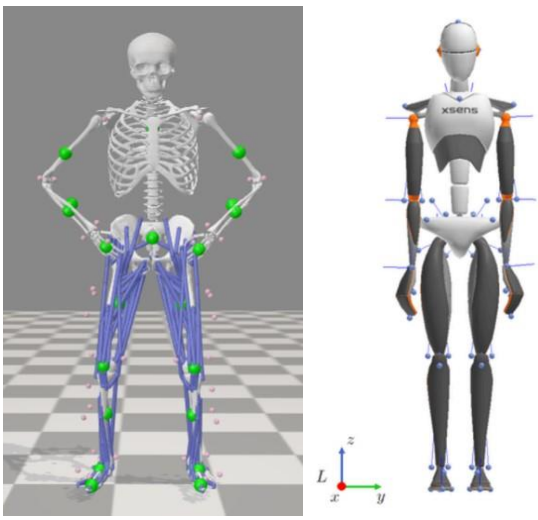


*Figure 19. Biomechanical models recreated by OpenSim and Xsens Analyze starting from the acquired data*

After the extraction of the markers' positions, it was fundamental to plot them in a proper reference system: this allowed to visually control the good acquisition of the jump for each trial and each subject, and also to evaluate significative differences between the recordings with the two different system of acquisitions (i.e. movements of the opposite limbs in the same time instants). To properly plot the markers, the consideration of the same reference system was fundamental. Since Xsens adopted as reference system the conventional one, no further elaborations needed to the be done on its data. Instead, for OpenCap some proper rotations had to be performed to report the positions of the markers in the same conventional reference system, starting from a frame mostly used in Anglo-Saxons derived software. The elaboration of the markers' coordinates to perform the right rotations had to be done for each acquired point.

Eventually, the synchronization of the signals from the different acquisition systems was performed. It was realized starting from the identification of the maximal position reached by the left shoulder's marker in each jump for both OpenCap and Xsens, and then by properly cutting the signals to have the same acquisition's length and to both contain all the jump-phase. This task was primarily performed with a visualization objective: the aim was to synchronize the signals to contemporaneously plot the jump phase and evaluate their difference evaluable with a visual look. Some attempts were made to evaluate the differences between the two acquisition systems in the spatial tracking of each single marker, but none could be performed with valid results. Indeed, two were the main reasons for this unsuccessful comparison. The first was the fact that the two acquisition systems were based and referred to completely different anatomical models, with different placement of the (then extracted) markers. The second was the presence of *spatial-fluctuations* of the acquired markers in both the two systems' acquired signals: these were due to some erroneous identification from the software of the positions of anatomical markers in some different time instants, or to some erroneous calculations of the body segments' orientations. Neither of these two problems allowed the perfect synchronization of the markers' time-signals, hence their proper comparison.

### 4.2.2 Extraction and computation of the centers of mass of most important body segments

The second task completed was the extraction or the computation of the *centers of mass* (CoMs) of the most important body segments. The unique way to estimate the ground reaction forces was to relate them to the forces applied on each body segments' center of mass, thus it was necessary to compute their positions in space. Two different procedures were used for OpenCap and Xsens acquired signals, because of their different anatomical models.

Regarding OpenCap, the downloaded files were opened and elaborated with the dedicated software OpenSim, which permitted to perform a scaling of the standard model on the typical anatomical parameters of each subject, to do the inverse kinematic analysis and finally to do also the inverse dynamic analysis (useful to find the internal forces and moments produced by the muscles). The first two passages were already completed by the online software connected to OpenCap, from which we correctly extracted the already performed kinetic analysis with the relative angles as output. The third possible analysis to be done with this dedicated software was not in the interest of the study, mainly for the absence of a validation system to verify its correctness and accuracy. In addition, from OpenSim it was possible to visualize the CoMs of the most important body segments in its adopted anatomical model (LaiUhlrich2022 model). Unfortunately, it was not possible to extract the time-series that consist in their positions for each time instant during the performed jumps.

Thus, to extract the CoMs of some of the most important segments of the human body a different anatomical model present in literature was considered: it provided not only the proximal and the distal joints that define each body segment of human body, but also all the masses (relative with respect to the total body mass) and the positions of the CoMs of those body segments (as a distance from the proximal joint along the line that connects the proximal and distal joints). The adopted model is the Winter's model, which is presented in Appendix.

After the choice of the anatomical reference model, another problem had to be faced and solved: OpenCap gave the positions of some anatomical markers that were not exactly the proximal and distal joints reported in the Winter's model. To extrapolate these joints' positions in each time instant of each jump, a first visual analysis of the markers acquired from OpenCap was done opening the downloaded files on OpenSim. From this visual analysis, only the markers that were connected and that defined the positions of the joints of interest for the study were considered. Then, the computation of the position of each joint for each recorded time instant was performed. Since the positions of the markers were given in a tri-dimensional space, to extract the correct joints positions a hypothesis was required: the wanted joint positions were placed at half of the straight line that connected two (or more, with an iterative procedure) recorded points related to each single joint. This procedure was done only for the few joints whose position was not directly extracted as a time-series from OpenCap (i.e. ulnar styloid).

Once the definition of the position of each joint needed to extract the CoMs of the body segments was completed, the computation of the placement in space of each of those centers of mass was performed. Since it was known the position of each CoM with respect to its proximal and distal joint, the adopted hypothesis was that this interest point was on the line that connected these joints positions: thus, the same procedure of before was adopted, with the unique difference of the different distance of the final extracted point from the proximal joint position. Indeed, the body segments do not have a homogenously distributed mass, neither a CoM which is not perfectly in the middle between their proximal and distal joints, but they are characterized by a more concentrated portion of mass near one or the other joint. By using the Winter's model, it was possible to evaluate this non-homogeneous mass distribution of each body segment, and to report it to a precise placement of their CoM in a precise position with respect to segment's relative joints. Since the Winter's model defines the position of the same center of mass even with respect to different joints, some estimated masses (reported as fraction of the total body mass M) are different for the same body segment, depending on the reference joints from which this segment is defined. As a consequence, in order to correctly distribute the total body mass on the different body segments' CoMs, it was necessary to modify the mass dedicated to the *Trunk, Head, Neck* body segment: it was no more 0.578M as in Winter, but 0.436M, which was the only one that summed to the other contributes of mass of the other body segments gave exactly the total body mass of the subject. Nevertheless, this was a reasonable hypothesis, as highlighted by the similar value of the total body mass portion given to this specific body segment in Winter's model.

Having described the passages to extract the body segments' CoMs from the time-series signal from OpenCap, it is necessary to describe the extraction of them from the signals acquired with Xsens.

Xsens can provide as outputs the position of each body segment's center of mass, allowing to avoid all the computations done for the OpenCap's signals. Moreover, the same anatomical model was used to obtain comparable results between the two systems.

### 4.2.3 Estimation of the GRF profiles and extraction of the parameters/variables of interest

The third fundamental passage performed in the analysis was the estimation of the GRF profiles and the extraction of the parameters and the variables of interest during jumping activities. To perform this specific task, a similar analysis to the one presented by Burton [11] was conducted.

Once the extraction of the centers of mass of each body segments of interest was performed for each acquisition system's signals, their positions time-series (both the vertical, the lateral and the frontal) were filtered at 4 Hz, using a third order Butterworth filter, before differentiating with respect to time to calculate the vertical, the lateral and the frontal velocities respectively. Segments velocities were again filtered at 4 Hz and differentiated over time to get the vertical, lateral and frontal segmental accelerations.

Each segmental acceleration was multiplied by its proper segment mass and summed to provide an estimation of the vertical, lateral and frontal GRF profiles, according to the formulas presented below:

$$GRF^{est} = \sum_{j=1}^{N} m_j * (a_j + g)$$

Where $GRF^{est}$ is the GRF estimated from segmental kinematics, $m_j$ and $a_j$ are the mass and the acceleration of the segment $j$ on the three directions respectively, and $g$ is the gravitational acceleration (equal to 9.81 m/s$^2$).

Accelerations of each segment, rather than the whole-body CoM, were used to allow future investigation of individual segmental contributions to GRF variables of interest, and to investigate if some limb asymmetries in the generation of forces could be evaluated from the segmental different GRF contributions. Forces were normalized to each athlete's body weight to allow their better and more direct representation. In Figure 20 are represented the profiles of the vertical components of the GRF during a DJ, both for OpenCap and Xsens.

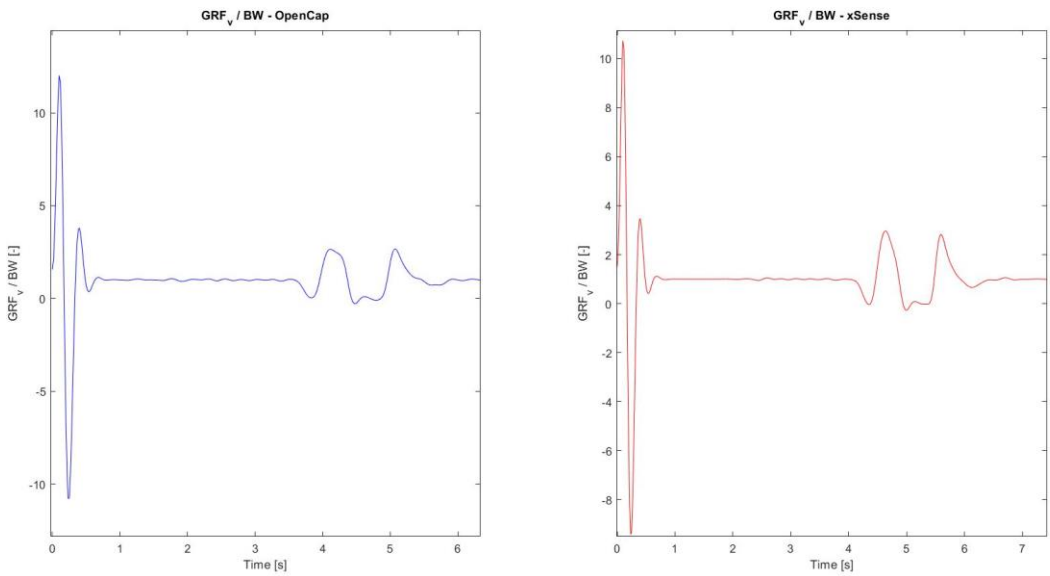


*Figure 20. OpenCap (left) and X-Sens (right) SJ's total GRF vertical components profiles.*

Although the estimation of the performances during a jump was done on the basis of the only whole-body vertical GRF profile, an additional analysis of the two different contributes to the GRF coming from the two different limbs was conducted, but with no considerable results (probably due to the adopted upper limb weights *equal-distribution* hypothesis).

As said previously, the computation of all the three components of GRF related to each inferior limb was performed. This choice allowed firstly to have a better graphical representation, by plotting these forces as the components of the force vector on the tri-dimensional representation of each jump activity. In addition, it could be seen as a great innovation, because it allowed the visual detection of the good performance of each jump attempt, by highlighting eventually wrong propulsive phases in the first part of the DJ, or some eccentric components during the SJ. One frame of the dynamic tri-dimensional representation of the DJ activity with the associated force vectors is reported in Figure 21.

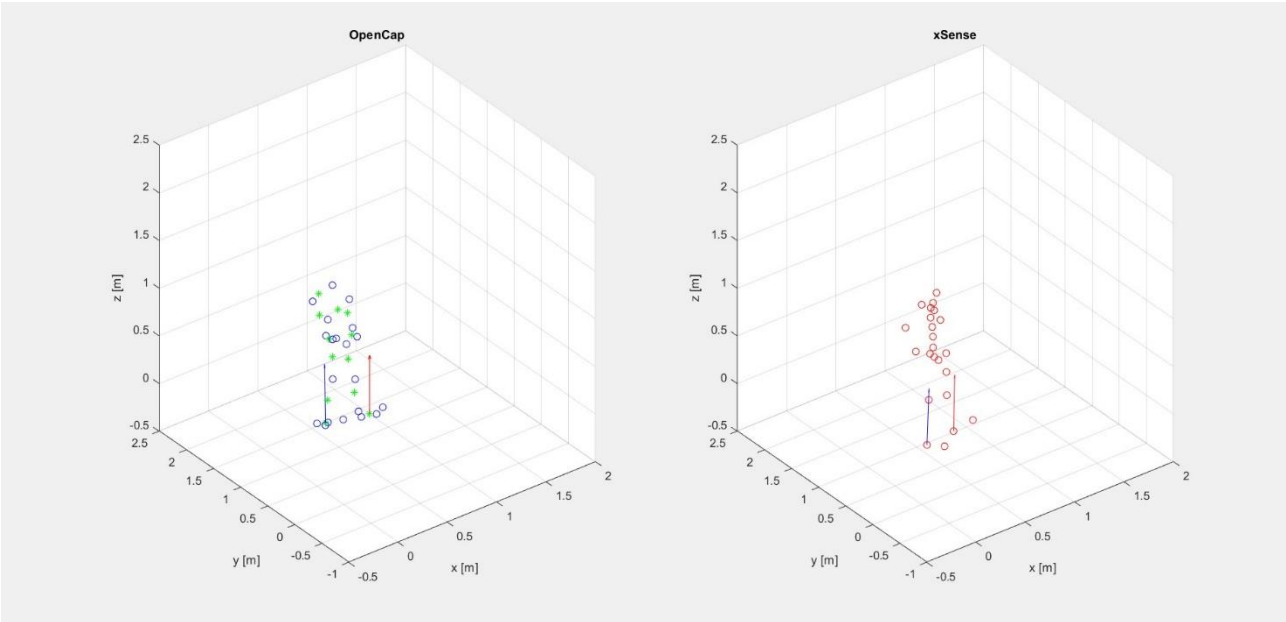


*Figure 21. OpenCap (left) and X-Sens (right) DJ's dynamic tri-dimensional representation.*

One important fact to highlight (also evident from the GRF profiles reported in the previous pictures) was the presence of some erroneous values of forces, as the two first elements of their time-series. These were due to the double differentiation of a finite series of position values, which were differentiated with MATLAB® function *gradient()*, that computed the derivative with the incremental ratio defined as the ratio between the difference of consecutive positions over the difference of the related consecutive time instants. For each differentiation, the length of the final vector was decreased by one, and the function *gradient()* overcame this different length problem by adding a zero value in the initial position of the new output-vector: these two zero values caused two erroneous values in the acceleration (and consequently in GRF) time-series. This was one of the main reasons that led to take into consideration new methods (and anatomical models) to estimate the GRF from Xsens, which is after described (see Appendix).

Consequently, force and kinematic variables were determined for the propulsive and landing phases (i.e. duration, peak/mean force and power, impulse, maximal velocity and acceleration, time to take-off/stabilisation), and for the jump (i.e. flight time, jump height from three calculation methods, reactive strength index (RSI) and reactive strength index modified (RSI-mod), excursion of the pelvis). In the following paragraphs not all the extracted parameters are described, but only the height of jump and the RSI, with their relative computation procedures. Nevertheless, a description of the take-off and landing instants, and of the starting time-instants of the eccentric and concentric phases must be presented, because they were fundamental for the computation of the two parameters of interest.

*Take-off and landing instants*

From the graph of the total vertical component of the estimated GRF, the take-off and the landing instants were extracted respectively as the ones where the ground reaction force reached the zero value after the jump concentric phase, and as the point where the force drastically increased (representing the impact with the ground).

*Starting instants of the concentric and eccentric phases*

From the graph of the vertical component of the total GRF, firstly the starting instant of the concentric phase was extracted: it was the one in correspondence of the last body weight (BW) value (*1* in the normalized graph) assumed by the GRF previous its sudden increase right before the take-off. Only after, the starting point of the eccentric phase was extracted: it was the point where the profile of the GRF started to move away from the value of the BW of the subject (so from the value of *1* in the normalized graph). From these extracted time instants, the durations of the concentric (from its starting point to the take-off instant) and of the eccentric (from its starting point to the concentric phase's one) phases were computed too.

*Height jump*

The height jump is one of the most important parameters in jump performances evaluation: this led to extrapolate it with three different methods. The first one consisted in extracting the height of the jump from the Z component of the marker on the pelvis position: it defined it as the difference between the maximal position reached by the marker in the flight phase and the vertical position at take-off. The second took from the same graph the maximal vertical position reached by the pelvis' marker and the position of the same marker at landing instant and managed to extract the height of the jump from their difference. The last procedure used to compute this parameter starting from the flight time: the duration of the flight phase was calculated as the difference between the landing and the take-off instants, then this was divided by 2 (hypothesizing an equal time spent in the upward and downward movement during the flight phase), and eventually it was elevated at the power of 2 and multiplied by the gravitational acceleration "g" (in accordance with the equations of a parabolic motion of a *falling body*). Finally, since the extrapolation of the maximal vertical position of the pelvis' marker was performed both manually and automatically (using the proper MATLAB® function), the computed jump height parameters were five for each trial.

*Reactive Strength Index (RSI) and Reactive Strength Index modified (RSI-mod)*

Starting from the previously computed height of the jump and from the durations of concentric and eccentric phase, the RSI and RSI-mod indexes were computed as the ratios between the height of the jump and the ground contact time or the time to take-off respectively. The two parameters assumed the same value, since from the investigation the ground contact time and the time to take-off could

be only evaluated as the same entity and the same sum of the durations of the concentric and eccentric phases. This problem was not relevant, because in normal investigations the RSI is an index properly referred to DJ activities, instead for the other type of jumps (which are without a proper eccentric phase) the RSI-mod is evaluated.

### 4.2.4 User-friendly data representation and Pearson's statistical analysis

The successive step was the implementation of a specific MATLAB® code able to collect the previous descripted results and to represent them with a user-friendly interface. To complete this task, it was decided to consider separately the results of the SJ and DJ tests for each subject, and to represent all the values recorded for each of those parameters with a *bar-graph*. The final result was really similar to the user interface of some software made by force plates' producers (i.e. Vald®), and it was a histogram graphic representation of each jump-parameter trial. In Figure 22 the results related to height jump of Subject 5's SJ and DJ performances are presented.

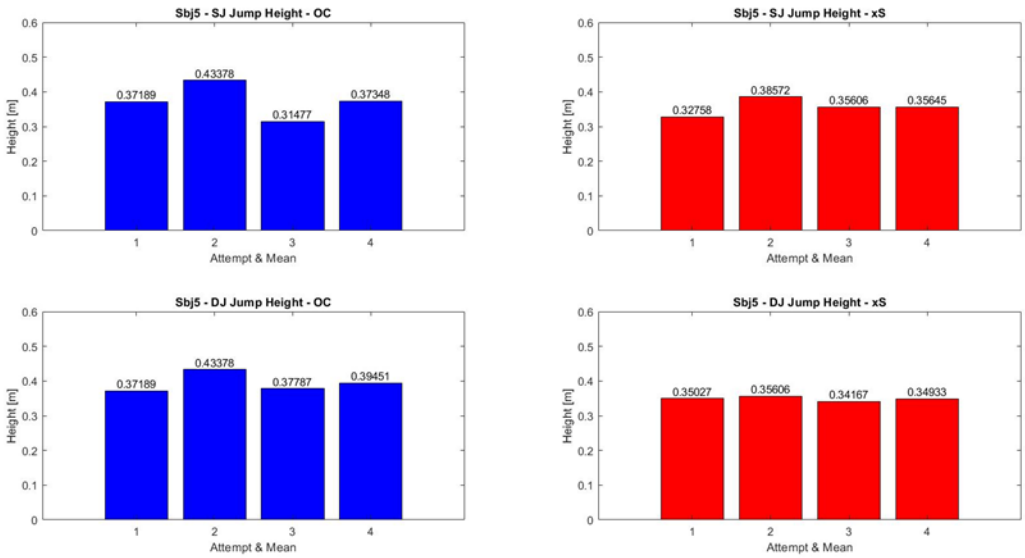


*Figure 22. OpenCap (left) and X-Sens (right) Subject 5 SJ and DJ's jump height results*

As following step, since the high mole of data recorded, a statistical analysis of the results was required. The study of the correlation of each subject's jump-parameters with other jump-parameters and with subject's features was conducted. This type of data evaluation could show possible links between recorded results and evaluate the dependencies of the most important parameters on some others' assumed values.

As a consequence, a Pearson's (or a Spearman's) statistical analysis was chosen to evaluate the presence of possible correlation among results. After having extracted all the above presented jump-parameters, the mean values of those related to the same subject and to the same jump-test were computed: at the end, for each subject's parameter, four mean values were extracted, two that were the average values of the parameter in SJ and DJ tests acquired with OpenCap, and the others that were the same but relative to the acquisition with Xsens. Then, for each parameter, four vectors containing all the mean values of the same test and related to all the different subjects were created, and the same was done for other subject's features, like the height, the weight, the BMI (Body Mass Index) and the Peak Power (extracted as the maximum among all the values of the products between

the GRF and the velocity in each time instant). Every one of those vectors represented one statistical population, which needed to be assessed as a *normal distribution* (with a Wilcoxon test) before performing the last data analysis with Pearson's test (in case of non-normal distribution, the alternative Spearman's test had to be used). These tests were both performed with MATLAB® already implemented *corr()* function, which gave as results the correlation coefficient (β) between the two input populations, followed by its *p-value*; if this value was lower than the conventional 5% ($p<0.05$) the correlation coefficient was called statistically significant and the two input populations were actually correlated.

In the following Table 9 the Pearson's (or Spearman's) analysis results are presented for the statistical correlations tests performed among the height of the jump, and another subject's feature, which is the Peak Power exerted in the phases before the jump.

*Table 9. OpenCap's statistical analysis results for SJ and DJ*

| Jump Height | 2nd population | β - SJ | *p-value* - SJ | β - DJ | *p-value* - DJ |
|---|---|---|---|---|---|
| From Flight Time | Subject's Peak Power | 0.9375 | 2.5256*e-7 | 0.7980 | 3.6274*e-4 |
| From Graph detection of Take-off | Subject's Peak Power | 0.8722 | 2.2414*e-5 | 0.8157 | 2.0950*e-4 |
| From Graph detection of Landing | Subject's Peak Power | 0.8986 | 5.3150*e-6 | 0.7001 | 0.0037 |
| From Function detection of Take-off | Subject's Peak Power | 0.8755 | 1.9049*e-5 | 0.8087 | 2.6219*e-4 |

*Table 10. Xsens' statistical analysis results for SJ and DJ*

| Jump Height | 2nd population | β - SJ | *p-value* - SJ | β - DJ | *p-value* - DJ |
|---|---|---|---|---|---|
| From Flight Time | Subject's Peak Power | 0.9208 | 1.1280*e-6 | 0.6788 | 0.0054 |
| From Graph detection of Take-off | Subject's Peak Power | 0.8812 | 1.4222*e-5 | 0.7485 | 0.0013 |
| From Graph detection of Landing | Subject's Peak Power | 0.8525 | 5.4147*e-5 | 0.5632 | 0.0288 |
| From Function detection of Take-off | Subject's Peak Power | 0.8799 | 1.5270*e-5 | 0.6315 | 0.0116 |

### 4.2.5 Validation of OpenCap by Xsens

Since the main aim of the study was the validation of the OpenCap acquisition system with Xsens, a final step was performed. Indeed, a comparison between the values of all the jump-parameters previously described and found from OpenCap and Xsens was completed.

The jump parameters were extracted, divided in two groups related respectively to SJ and DJ tests, and, for each trial of each jump-test the difference between the parameter's values estimated from OpenCap and from Xsens was computed; this was done for each subject and for each test. After this initial difference, also the absolute modulus of every difference was performed. With these simple calculations it was possible the evaluation of the errors and of the mean absolute errors (MAE), fundamental for the validation of OpenCap taking as reference Xsens, like what was done for the kinematic analysis presented above.

After having computed the errors and the MAE, the calculation of their average values and standard deviation was conducted [11]. This allowed to evaluate the mean error that OpenCap did (compared with the already validate Xsens system) for each specific jump and for each different subject (some errors could be related to the only execution of the jump or to the body shape of the subject), but also the range of values in which this error could shift.

In Figure 23 the mean error values and the relative standard deviations for some of the most important jump-parameters for SJ and DJ tests are presented. In these graphs it is clear that every MAE must be interpreted with their relative unit of measure, which is the one of the parameters evaluated to compute each of them (as presented in the graph).

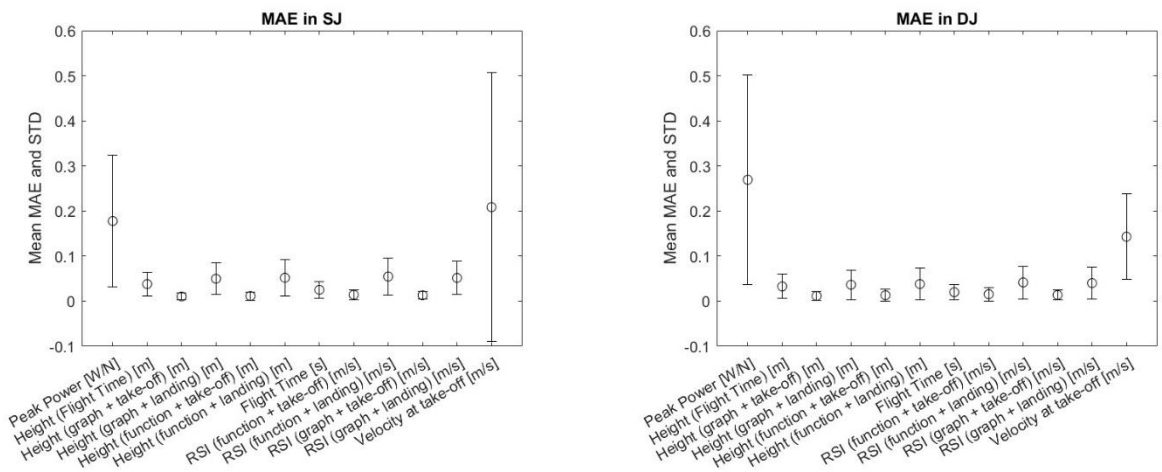


*Figure 23. OpenCap (left) and X-Sens (right) MAEs and STDs for several jump-parameters*

### 4.2.6 Discussion of the results

Regarding the statistical analyses conducted for the acquired data, some remarkable conclusions could be done. Firstly, the presented results only regarded the correlation between the different extracted jump's heights and the Peak Power exerted by the subject during the jump phase. These were the ones that showed the best inter-relation between the two tested statistical populations: in *Table 10. Xsens' statistical analysis results for SJ and DJ*Table 10 the respective p-values were drastically lower

than 5% for the parameters extracted from OpenCap (2.5256e-7 – 2.2414e-4 for SJ; 2.0950e-4 – 0.0037 for DJ), and considerably below this value for Xsens's ones (1.1280e-6 – 5.4147e-5 for SJ; 0.0013 – 0.0459 for DJ). Instead, for other tested parameters the statistical analyses did not show their effective influence in the determination of the height of the jump. In addition, focusing on the comparison between the results of the two acquisition systems, the presence of a statistical correlation between the same populations for both the systems sustained the thesis of the identification of OpenCap as a valid MoCap alternative to Xsens and to an optoelectronic system.

On the base of the values assumed by the MAEs between OpenCap and Xsens, an effective validation of the experimental acquisition system could be conducted. By dividing the presented results in three main groups, some conclusions could be made.

Regarding the jump's heights extracted from both the systems, the mean values of their MAEs were 0.03234 [m] for SJ's trials and 0.02648 [m] for DJ's ones, and their maximal standard deviations were 0.0400 [m] for SJ and 0.0349 [m]. These results showed that the absolute error made by OpenCap was in the range of 3 [cm]: this intrinsic error was significant, especially if we consider jump's heights that ranged from 10 to 35 [cm], but it was acceptable considering OpenCap is a completely marker-less acquisition system, based only on smartphones' cameras acquisitions.

Looking at RSIs' MAEs, their mean values were 0.03347 [m/s] for SJ's trials and 0.02783 [m/s] for DJ's ones, with a maximal standard deviation were 0.0409 [m/s] for SJ and 0.0362 [m/s] for DJ. This showed that even in the evaluation of the RSI indexes the two acquisition systems showed pretty similar results, with an error that must be considered but that was not considerably high.

Finally, the maximal MAEs (from both SJ and DJ trials) related to other parameters like Peak Power, Flight Time and Velocity at take-off were respectively 0.2692 [W/N], 0.0252 [s] and 0.2085 [m/s], with correspondent standard deviations of 0.2319 [W/N], 0.0178 [s] and 0.2983 [m/s].

In conclusion, the presented results allowed the effective validation of OpenCap considering Xsens as gold standard. The estimation of the most important jump's performances and parameters (jump's height and RSI) was possible with this new acquisition system with acceptable MAEs. These errors were considered acceptable because of the revolutionary complete marker-less evaluation of the motion of OpenCap, and of the relative additional errors due to manual extraction of the parameters and to the double derivation of discrete acquired data. Eventually, regarding the dynamic task's analysis, OpenCap could be considered a valid alternative to the other marker-less acquisition systems.

# 5. Conclusion

The results obtained by the several methods used for the kinematic analysis showed an overall congruence. The two approaches applied for the calculation of the mean absolute error (MAE) brought to the conclusion that the OpenCap was able to acquire the same angles as the Xsens for the knee and ankle joint with a mean MAE of 5°, while concerning the hip joint, the error reached a higher mean of around 20°. Certainly, the value found for the hip showed the presence of some complications or disturbances in the acquisition of the OpenCap system: the clothing or the placement of the hands at the level of the hip could had participated in the difficulty of the detection of the joint by the marker-less system. On the other hand, the mean error of 5° could be considered acceptable to validate the OpenCap. This result contained inside also the uncertainty introduced

using the inertial sensors (wrong placement or movement due to soft tissues) respect to the gold standard marker-based motion capture systems. Related to the range of motion (ROM) approach, the Wilcoxon test between the two ranges detected by OpenCap and Xsens reported similar results to the previous methodologies: the best correlation was found for the ankle joint, while for the knee only half of the parameter were positive and for the hip all the results of the test gave back a negative correlation. It was possible to state that the validation of the marker-less OpenCap system for the kinematic analysis of human body's lower limbs could be considered as verified if related to knee and ankle flexion/extension in the sagittal plane. Concerning the validity for the hip flexion/extension in the same plane, further studies should be conducted to test the influence of external factor on the acquisition of the movements, like the presence of other body segments/parts near the hip or particular types of clothing.

Concerning the dynamic analysis, the results showed an overall congruence too. The several approaches performed for the calculation of the height of the jump (and the other most important parameters) brought to the conclusion that OpenCap was a valid alternative for its extraction with a mean MAE of 0.03234 [m] for SJ's trials and 0.02648 [m] for DJ's ones. Among the two evaluated tasks, the maximal standard deviation was about 0.0400 [m]. Different MAEs and STDs (relative to their related parameters' values) were computed for other parameters, like RSI and Peak Power. As previously presented, the values of the evaluated MAEs were affected by disturbances in OpenCap acquisitions (like the clothing or random fluctuation of the recreated anatomical markers) or due to the incorrect placement of the IMUs on the body segments for the Xsens' acquisitions. Nevertheless, both the MAEs' relative low values and the similar results of Pearson's and Spearman's correlations tests (between jump's height and different parameters) for the two acquisition systems stated the validation of the marker-less OpenCap system also for the dynamic analysis of human body during jumping tests. Further studies should be conducted to improve the GRF extraction ability from OpenCap acquisitions and to estimate different limbs' contribution to the total GRF.

# Appendix

*Alternative dynamics analysis*

In this chapter a further analysis performed on SJ and DJ is presented. It focuses on the data coming from Xsens. The final aim of this appendix was to compute the ground reaction forces (GRF) exhibited during these activities with a model more suitable for the Xsens' body segments. Moreover, the correlation between some parameters and the height of the jump was investigated. Indeed, the height of the jump is what is usually measured to evaluate the athlete's performance.

IMUs provided several data but in the following analysis only the kinematic ones (positions, velocities and acceleration of human body segments and of the global centre of mass) were utilised. In this case the human body model used was the Erdmann's one because it was the most suitable for the human body segments given by the Xsens. The distribution of mass is reported in the following Table.

*Table 11: Mass distribution of human body segments (Erdmann Model)*

| *Segment* | *Singular [%]* | *Double [%]* |
|---|---|---|
| Head | 5 | 5 |
| Neck | 3 | 3 |
| Thorax | 11.6 | 11.6 |
| Abdomen | 16.4 | 16.4 |
| Pelvis | 11.4 | 11.4 |
| Shoulder | 5.7 | 12.4 |
| Arm | 2.6 | 5.2 |
| Forearm | 1.5 | 3 |
| Hand | 0.6 | 1.2 |
| Thight | 10 | 20 |
| Shank | 4.5 | 9.0 |
| Foot | 1.4 | 2.8 |
| | Total | 100 |

The mass distribution on all the different body segments was necessary to compute the inertial and gravitational forces following the same procedure discussed before. From these, the GRFs were computed in the same way.

In this appendix the height of the jump has been computed in two different ways: firstly, considering the body's centre of mass and then starting from the flight time of the jump.
The first step, necessary for both the methods, was to define the take-off and the landing phase. They were selected by making some considerations on the GRF graph: when a subject started to detach from the ground, the ground reaction force showed a negative value in the graph. In principle while the subject was not touching the ground the GRF should be zero but, since it was not directly measured with a force platform, it showed a negative value just after the take-off. At the end, the first negative point was considered to be the take-off instant. This should not be considered exactly true but, since the sampling frequency of acquisition is quite high, the resultant error has been considered negligible. To determine the landing instant, similar considerations have been made. When a subject landed after the jump, the GRF force showed a peak, so the landing instant was selected as the first negative point before the peak of the GRF. The time instants were plotted both in the GRF and in the CoM graph as it's possible to see in Figure 24. The take-off and landing instants are the two last dots on the diagram; the function of the first two dots will be explained in the RSI section.

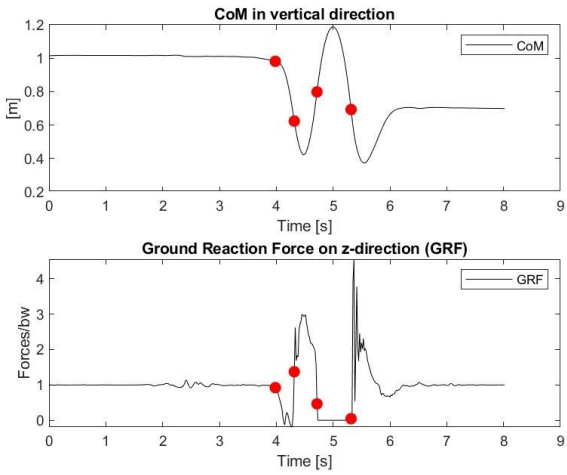


*Figure 24. CoM displacement and GRF during a squat jump*

The height of the jump was computed firstly applying the formula coming from the theory of fall of an object. This formula has some simplifications (such as not considering the resistance of the air) that have been considered reasonable. It is reported down below.

$$h_{jump} = 1{,}226 \cdot t_{jump}^2$$

The height was computed also considering the centre of mass. With this method the height was obtained by making the difference between the maximum of the whole-body centre of mass vertical displacement curve and the position of the same point at the take-off instant. The same was done also considering the position of CoM at the landing instant. Since the position of the take-off instant and the one at the landing instant were not exactly the same, the two results were averaged to get a more accurate estimation of the real height of the jump. The position of the centre of mass at take-off and landing instants can be observed in Figure 24. The RSI index was computed too. Since the

height of the jump was computed with two methods, also the RSI index was calculated separately considering the two different height's values.

### Statistical Analysis

The aim of the last part of the study was to assess which parameters could influence the performance of athletes in the squatting activities. In details, the influence of height and BMI of subjects, and of the maximum peak force had been studied to evaluate their correlation with the height of the jump. In the previous chapter, the relation between Peak Power and height of the jump showed a good result, in this case the aim was to see if the same could be said starting from the GRF. Once again, the Pearson's test was performed and, if the normal distribution hypothesis failed then the Spearman's test was consequently used, always with the *corr()* function of MATLAB®.

Now, the results are reported. As first, the height of the jump has been plotted considering the two methods used to compute it and the corresponding mean (*Figure 25. Height of the squat jumps and drop jumps*Figure 25).

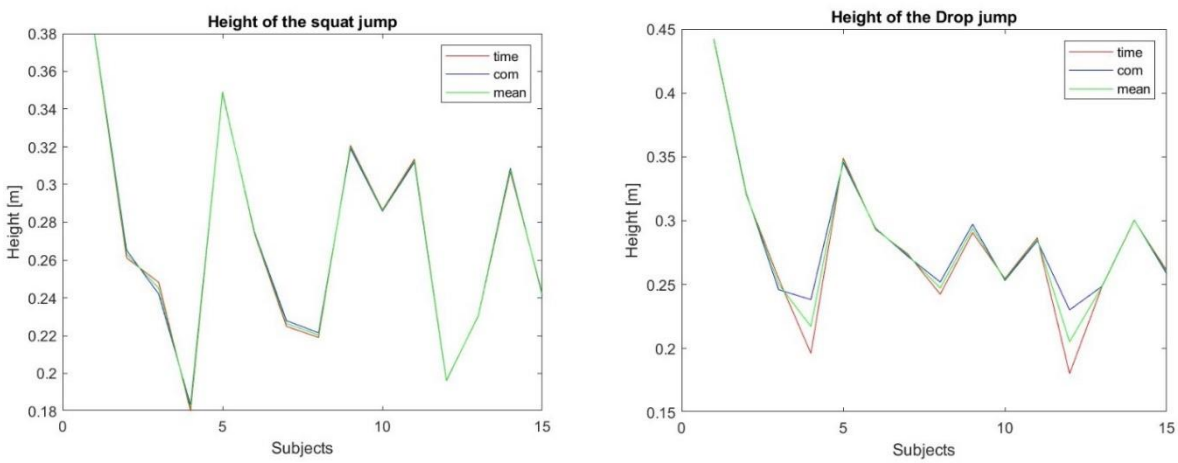


*Figure 25. Height of the squat jumps and drop jumps*

The data plotted were the ones obtained by averaging the three acquisitions performed by each subject and the mean was the average between the mean height obtained from CoM displacement and the one obtained by time of the jump. As it is possible to see, the two methods' results were really similar especially considering the squat jumps. In the drop jumps more variability could be observed. The same could be done considering the RSI (Figure 26).

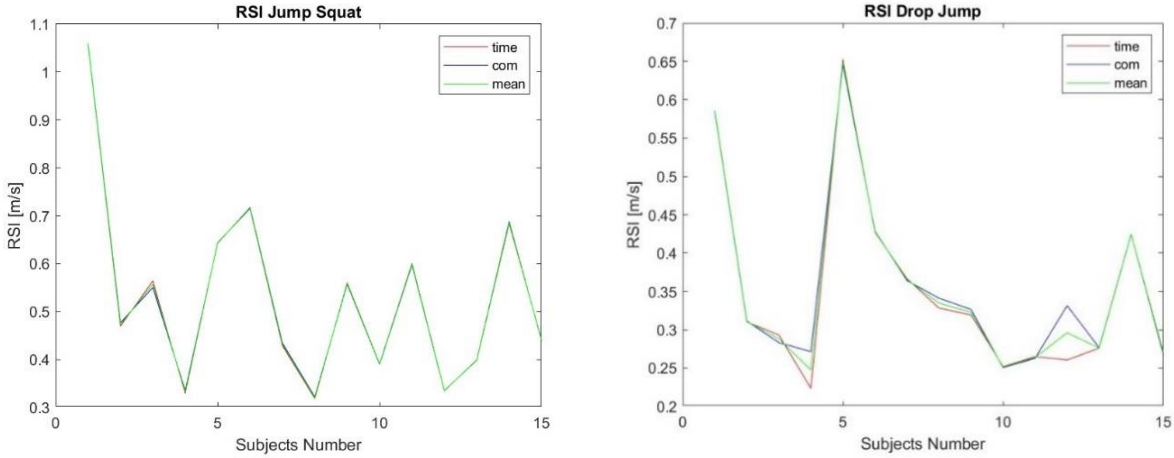


*Figure 26. RSI from squat jump and drop jumps*

Since this index depends on the height of the jump, the pattern was similar to the one observed before for the height of the jump.

Eventually, the statistical analysis was performed. Both the Pearson's and the Spearman's tests start from the null hypothesis of correlation between the two set of data. As a consequence, a *p-value* lower than 0.05 means that the data are strongly correlated, on the other hand, higher *p-values* indicate low or no correlation. The correlation had been investigated considering the height of the jump (computed with the two methods described before and the correspondent mean of their results) on one hand, and the BMI, height of the subjects and the Peak Force on the other hand. In this case the Peak Force was intended to be the one of the GRF before the take-off instant. The *p-values* are shown below in Table 12.

*Table 12. p-values from Pearson and Spearman tests*

| Variable | Height computation's method | Type of Jump | p-value |
|---|---|---|---|
| BMI | CoM | SJ | 0.636 |
| BMI | Time | SJ | 0.640 |
| BMI | CoM | SJ | 0.638 |
| BMI | Time | DJ | 0.859 |
| BMI | CoM | DJ | 0.996 |
| BMI | Time | DJ | 0.840 |
| Height | CoM | SJ | 0.405 |
| Height | Time | SJ | 0.392 |
| Height | CoM | SJ | 0.398 |
| Height | Time | DJ | 0.382 |
| Height | CoM | DJ | 0.348 |
| Height | Time | DJ | 0.412 |
| Peak of Force | CoM | SJ | 0.004 |
| Peak of Force | Time | SJ | 0.004 |
| Peak of Force | CoM | SJ | 0.004 |
| Peak of Force | Time | DJ | 0.903 |
| Peak of Force | CoM | DJ | 0.893 |
| Peak of Force | Time | DJ | 0.944 |

As it is possible to notice, the BMI and the height of the subjects were not correlated with the height of the jump. On the contrary, the Peak Force showed a very strong correlation with the height of the jump (no matter the way of computation used), but only considering the squat jumps. The drop jumps were not statistically correlated to any of the tested parameters.

## Conclusion

From the analysis performed, it was possible to notice how the height of the jump estimated from flight time and from the displacement of the centre of mass was really close in both the tasks investigated. However, in the drop jump activity higher differences could be seen, especially considering specific subjects like the number 4 and 12. Nevertheless, the differences stayed in the range of few centimetres that could be considered acceptable considering the task performed (it is more complex than the squat jump); remembering also that only three attempts were recorded and the results plotted before were averaged on them, a non-proper acquisition could highly impact on the final results. Regarding the differences in the RSI's graph, it was clear that similar conclusions could be taken out since the RSI was computed starting from the height of the jump. Furthermore, in the RSI's graph the most important observation that could be made was that the RSI generally showed higher results in the squat jump activity. This was a consequence of the fact that it was a pure concentric task, so the time considered in the computation of RSI was shorter (no eccentric phase). In literature, the RSI index is usually linked to muscle stiffness of the subjects. In this study it could be observed that higher values of RSI are the ones related to subjects that usually practice sports where muscle stiffness is a key element.

Considering the statistical analysis, the BMI resulted to be non-correlated to the height of the jump. This could be considered a little bit counterintuitive, but it could be explained considering the ratio between the fat mass and lean mass: both contribute to the overall mass of the subject, but higher percentages of fat mass could prevent the subject to jump as high as someone with the same mass but lower fat mass. In this case the percentage of fat and lean mass was not evaluated, so this could be a future development.

The height of the subjects had a *p-value* that ranges from 0.382 and 0.412 that statistically indicated a weak correlation with the jump's height. This should be evaluated more in details since the sample of this study was limited to only 15 people.

The peak of force resulted to be well correlated with the height of squat jumps (Figure 27), but not with the one of drop jumps.

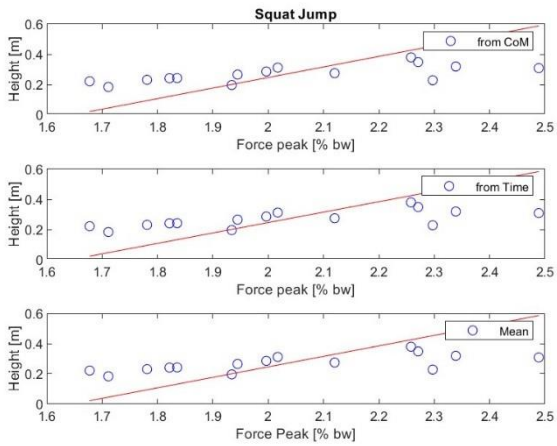


*Figure 27. Correlation between force peak height of the jump in Squat Jumps*

In principle, the expectation was to find a correlation between the force peak and jump's height for both the tasks. The non expected *p-values* of the drop jumps were probably related to the fact that the height of the jump could be more influenced by the power rather than the force. Indeed, comparing the ground reaction forces of two subjects that jumped different heights such as number 5 and 12, a higher peak of force could be observed in subject 12, even though the height of the jump resulted to be well lower compared to the one of subject 5 that had a lower peak of force.

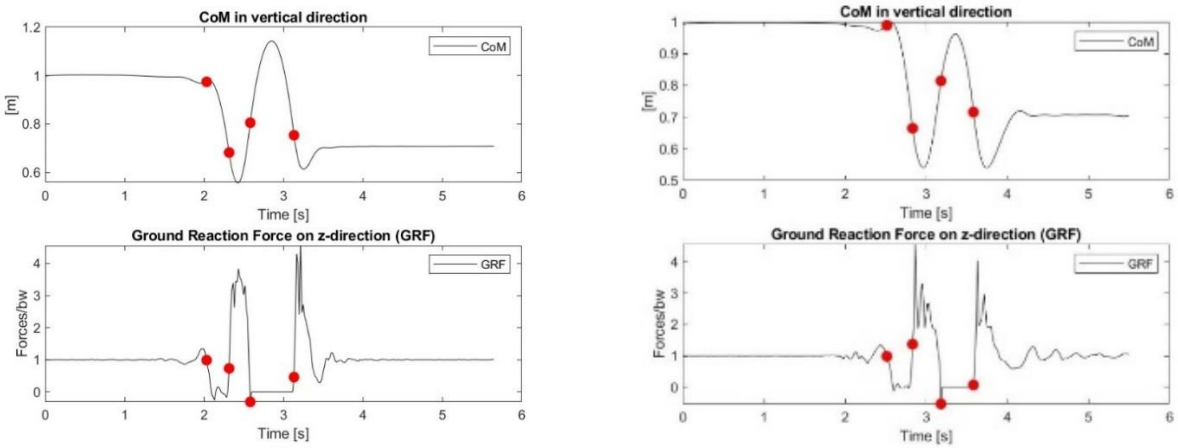


*Figure 28. GRF of subject 5 (left) and subject 12 (right)*

Further studies concerning the relationship between power (considering both the peak and the mean power) and the height of the jump could be performed to better characterize their influence on jumping performances.

*Winter's model*

In Table *11* are reported the list of the extracted centers of mass, with the respective proximal and distal joints from which they were extracted, their distances from one joint and from the other, and with their concentrated and representative mass.

*Table 9: Anthropometric data from Winter's model*

| Segment | Definition | Segment weight / Total body weight | Centre of mass / Segment length | |
|---|---|---|---|---|
| | | | Proximal | Distal |
| Left Upper Arm | Left Glenohumeral axis / Left Elbow axis | 0.028M | 0.436 | 0.564 |
| Right Upper Arm | Right Glenohumeral axis / Right Elbow axis | 0.028M | 0.436 | 0.564 |
| Trunk, Head, Neck | Greater trochanter / Glenohumeral joint | 0.436M | 0.66 | 0.34 |
| Left Forearm and Hand | Left Elbow axis / Left Ulnar styloid | 0.022M | 0.682 | 0.318 |
| Right Forearm and Hand | Right Elbow axis / Right Ulnar styloid | 0.022M | 0.682 | 0.318 |
| Pelvis | L4-L5 / Greater trochanter | 0.142M | 0.105 | 0.895 |
| Left Thigh | Left Greater trochanter / Left Femoral condyle | 0.1M | 0.433 | 0.567 |
| Right Thigh | Right Greater trochanter / Right Femoral condyle | 0.1M | 0.433 | 0.567 |
| Left Shank | Left Femoral condyle / Left Medial malleolus | 0.0465M | 0.606 | 0.394 |
| Right Shank | Right Femoral condyle / Right Medial malleolus | 0.0465M | 0.606 | 0.394 |
| Left Foot | Left Lateral malleolus / Left Head metatarsal II | 0.0145M | 0.5 | 0.5 |
| Right Foot | Right Lateral malleolus / Right Head metatarsal II | 0.0145M | 0.5 | 0.5 |

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